

Formation of Differential Equations:

The differential equations can be formed by differentiating the ordinary equation and eliminating the arbitrary constants.

Ex. 1. Form the differential equation by eliminating arbitrary constants, in the following cases and also write down the order of the differential equations obtained.

(a) $y = Ax + A^2$

Solution: $y = Ax + A^2 \rightarrow (1)$

on differentiation

$$\frac{dy}{dx} = A$$

Putting the value of A in equation (1), we get

$$y = \frac{dy}{dx} x + \left(\frac{dy}{dx}\right)^2$$

Answer:

on eliminating one constant A we get the differential equation of order 1 and degree 2.

(b) $y^2 = Ax^2 + Bx + C$

Solution $y^2 = Ax^2 + Bx + C \rightarrow (1)$

on Differentiation

$$2y \frac{dy}{dx} = 2Ax + B \quad \text{or} \quad B = 2y \frac{dy}{dx} - 2Ax$$

Again Differentiation

$$2 \left[\frac{dy}{dx} \cdot \frac{dy}{dx} + y \frac{d^2y}{dx^2} \right] = -2A$$

$$\text{or} \quad A = \left(\frac{dy}{dx}\right)^2 + y \frac{d^2y}{dx^2}$$

Again Differentiation w.r to x we get

(b) $y = A \cos x + B \sin x \rightarrow (1)$

Sol. on differentiation

$$\frac{dy}{dx} = -A \sin x + B \cos x$$

Again Differentiating

$$\frac{d^2y}{dx^2} = -A \cos x - B \sin x$$

$$\text{or} \quad \frac{d^2y}{dx^2} = - (A \cos x + B \sin x)$$

From equation (1)

$$\frac{d^2y}{dx^2} = -y$$

$$\text{or} \quad \boxed{\frac{d^2y}{dx^2} + y = 0} \quad \text{Answer}$$

This is differential equation of order 2 obtained by eliminating two constants A and B.

PE # 02

R $\left(\frac{dy}{dx}\right)^2 + y \frac{d^2y}{dx^2} = A$

$$2\left(\frac{dy}{dx}\right) \cdot \frac{d^2y}{dx^2} + \frac{dy}{dx} \cdot \frac{d^2y}{dx^2} + y \frac{d^3y}{dx^3} = 0$$

$$2 \frac{dy}{dx} \cdot \frac{d^2y}{dx^2} + \frac{dy}{dx} \cdot \frac{d^2y}{dx^2} + y \frac{d^3y}{dx^3} = 0$$

$$\boxed{y \frac{d^3y}{dx^3} + 3 \frac{dy}{dx} \cdot \frac{d^2y}{dx^2} = 0} \quad \text{Answer}$$

This is the differential equation of order 3, obtained the eliminating three constants A, B, C.

$y = ax^2 + bx + c$

There are three arbitrary constants so we can differentiate above equation 3 time.

eg. $y = ax^2 + bx + c$

on differentiation

$$\frac{dy}{dx} = 2ax + b$$

again Differentiation

$$\frac{d^2y}{dx^2} = 2a$$

now again Differentiation

$$\boxed{\frac{d^3y}{dx^3} = 0} \quad \text{Answer}$$

This is the D.E of order 3

So we do 4 times differentiation - Given equation

on differentiation

$$\frac{dy}{dx} = ae^x - be^{-x} - c \sin x + d \cos x$$

Again D.w.r to x

$$\frac{d^2y}{dx^2} = ae^x + be^{-x} - c \cos x - d \sin x$$

Again D.w.r to x

$$\frac{d^3y}{dx^3} = ae^x - be^{-x} + c \sin x - d \cos x$$

now last time D.w.r to x

$$\frac{d^4y}{dx^4} = ae^x + be^{-x} + c \cos x + d \sin x$$

$$\frac{d^4y}{dx^4} = y$$

$$\boxed{\frac{d^4y}{dx^4} - y = 0} \quad \text{Answer}$$

This is required D.E of order 4

(c) $y = ae^x + be^{-x} + c \cos x + d \sin x$

Above Given equation have involve four arbitrary constants.

(f)

$$y = A \cos 2t + B \sin 2t$$

Two arbitrary constants involve in above equation so we can apply + two times derivative.

$$\frac{dy}{dx} = 2A \sin 2t + 2B \cos 2t$$

Again

$$\frac{d^2y}{dx^2} = -4A \cos 2t - 4B \sin 2t$$

$$\frac{d^2y}{dx^2} = -4(A \cos 2t + B \sin 2t)$$

$$\frac{d^2y}{dx^2} = -4y$$

$$\boxed{\frac{d^2y}{dx^2} + 4y = 0}$$

Answer. 2nd order

(g) $(x-a)^2 + y^2 = a^2 \rightarrow (A)$

Diff. w. r to x.

$$2(x-a) + 2y \frac{dy}{dx} = 0$$

$$\text{or } 2y \frac{dy}{dx} = -2(x-a)$$

$$y \frac{dy}{dx} = -(x-a)$$

$$\text{now } y \frac{dy}{dx} - x = -a$$

$$\text{or } \boxed{a = x - y \frac{dy}{dx}}$$

Put this value in equation (A)

$$\left[x - \left(x - y \frac{dy}{dx} \right) \right]^2 + y^2 = \left(x - y \frac{dy}{dx} \right)^2$$

Answer.

(h) $y = e^x (C_1 \sin x + C_2 \cos x)$

Solution by C.V formula.

$$y' = e^x (C_1 \sin x + C_2 \cos x) + e^x (C_1 \cos x - C_2 \sin x)$$

$$y' = y + e^x (C_1 \cos x - C_2 \sin x)$$

$$\text{or } y' - y = e^x (C_1 \cos x - C_2 \sin x) \rightarrow (A)$$

Again D. w. r to x.

$$y'' - y' = e^x (C_1 \cos x - C_2 \sin x) + e^x (-C_1 \sin x - C_2 \cos x)$$

$$\text{or } y'' - y' = e^x (C_1 \cos x - C_2 \sin x) - e^x (C_1 \sin x + C_2 \cos x)$$

$$y'' - y' = e^x (C_1 \cos x - C_2 \sin x) - y$$

$$y'' - y' + y = e^x (C_1 \cos x - C_2 \sin x) \rightarrow (B)$$

By Putting eqn (A)

$$y'' - y' + y = y' - y$$

$$\text{or } y'' - y' + y' - y' + y = 0$$

$$y'' + 2y = 0$$

$$\text{or } \boxed{y'' + 2y = 0} \quad \text{Answer}$$

$$(I) \ y = \underline{a} \sin^{-1} x \rightarrow (1)$$

Solution

Differentiate w.r.to x

$$\frac{dy}{dx} = a \frac{d}{dx} (\sin^{-1} x)$$

$$\therefore \frac{d}{dx} (\sin^{-1} x) = \frac{1}{\sqrt{1-x^2}}$$

$$\frac{dy}{dx} = a \left(\frac{1}{\sqrt{1-x^2}} \right)$$

$$\therefore \frac{dy}{dx} = \frac{a}{\sqrt{1-x^2}}$$

now $\boxed{\frac{dy}{dx} (\sqrt{1-x^2}) = a}$

Put the value of a in equation (1).

$$y = a \sin^{-1} x$$

$$y = \frac{dy}{dx} (\sqrt{1-x^2}) \sin^{-1} x$$

$$\therefore y = \frac{dy}{dx} (\sqrt{1-x^2}) \times \frac{1}{\sin x}$$

$$y \sin x = \frac{dy}{dx} (\sqrt{1-x^2})$$

$$\therefore \frac{y \sin x}{\sqrt{1-x^2}} = \frac{dy}{dx}$$

$$\therefore \boxed{\frac{dy}{dx} = \frac{y \sin x}{\sqrt{1-x^2}}}$$

This is Required Answer.